

Danh gia du an RL Inventory Management theo tieu chi cua Thay

Ngày danh gia: 2026-05-27 Phien ban: v2 — Sau khi cai thien (20K episodes, them metrics, Dashboard UI)

1. Tong quan diem uoc tinh

#	Tieu chi	Trong so	Diem hien tai	Diem toi da	Trang thai
i	MDP dung, state/action/reward hop ly	20%	18/20	20	Thieu giai thich Markov trong bao cao
ii	Moi truong tu cai,kiem thu tot	15%	15/15	15	DAT
iii	Q-Learning va SARSA/Double Q dung	25%	21/25	25	RL > Random+AlwaysOrder2, chua thang Reorder
iv	Danh gia nhieu seed, baseline ro	20%	18/20	20	Da bo sung metrics
v	Demo chay duoc, truc quan, giai thich duoc	15%	12/15	15	Dashboard da verify, thieu video
vi	Ma sach, bao cao ro, tai lap duoc	5%	3/5	5	Thieu final_report.pdf
	TONG	100%	~87/100	100	Tang tu 76 len 87

So sanh tien bo

Hang muc	Truoc	Sau	Thay doi
Training episodes	8,000	20,000	+150%
Q-Learning profit	298.2	321.1	+7.7%
SARSA profit	316.5	339.6	+7.3%
Double Q profit	286.2	328.3	+14.7%
Missing metrics	3 thieu	0 thieu	Da bo sung
Dashboard UI	Emoji lon xon	Thong nhat indigo/slate	Chuyen nghiep
Daily reward chart	Khong co	Co	Moi

2. Danh gia chi tiet tung tieu chi

i. MDP dung, state/action/reward hop ly (20%) -> 18/20

DA DAT

Yeu cau	Code	Trang thai
State = (inventory, demand_regime, day_of_week, pending_order)	custom_env.py	DAT
2,646 states = 21 x 3 x 7 x 6	N_STATES = 2646	DAT
Action = {0,1,2,3,4,5} (6 actions)	N_ACTIONS = 6	DAT
Reward = revenue - purchase - holding - stockout	custom_env.py	DAT
Gia ban=10, gia nhap=5, ton kho=-0.5, stockout=-3	Cac hang so	DAT
Dynamics: pending ve dau ngay, cap 20	step()	DAT
Episode = 30 ngay	EPISODE_LENGTH = 30	DAT
Demand stochastic theo regime	DEMAND_DISTRIBUTIONS	DAT

CON THIEU

Thieu	Huong khac phuc
Giai thich Markov assumption trong bao cao	Viet: state hien tai chua du thong tin de ra quyet dinh. Demand la Markov (chi phu thuc regime).
Giai thich cong thuc Q-Learning trong bao cao	Giai thich alpha, gamma, TD target, TD error, terminal case. Code da dung.

ii. Moi truong tu cai, kiem thu tot (15%) -> 15/15 (HOAN HAO)

Yeu cau	Test	Trang thai
reset(seed)	test_same_seed_same_trajectory	DAT
step(action) tra ve 5 gia tri	test_step_returns_correct_format	DAT
render()	test_render_returns_string	DAT
state_encoder()	test_roundtrip_all_states, test_encoder_unique	DAT
state_decoder()	test_roundtrip_all_states	DAT
Test transition	test_pending_order_received_next_day, inventory_capped, never_negative	DAT
Test reward	14 test cases trong test_rewards.py	DAT
Test terminal	test_episode_terminates_at_30_days	DAT
Test boundary	test_zero_inventory, test_max_inventory, test_invalid_action	DAT
Test seed	test_same_seed, test_different_seeds, test_multiple_resets	DAT
Test encoder/decoder	8 test cases	DAT
KHONG dung Gymnasium	Tu viet BaseEnv abstract class	DAT

Ket qua: 50/50 tests passed trong 0.16s.

iii. Q-Learning va SARSA/Double Q dung (25%) -> 21/25

Thuat toan — TAT CA DUNG

Yeu cau	File	Trang thai
Random Agent	random_agent.py	DAT
Heuristic (AlwaysOrder2)	heuristic_agent.py	DAT
Heuristic (ReorderThreshold)	heuristic_agent.py	DAT
Q-Learning (off-policy)	q_learning.py — $Q[s,a] += a(r + g\max_{a'} Q[s',a']) - Q[s,a]$	DAT
Terminal: target = r	if done: td_target = reward	DAT
SARSA (on-policy)	sarsa.py — $Q[s,a] += a(r + gQ[s',a'] - Q[s,a])$	DAT
SARSA dung actual next action	td_target = reward + g * $Q[s', next_action]$	DAT
Double Q-Learning	double_q_learning.py — 2 bang Q1/Q2, cross-evaluation	DAT
Epsilon-greedy	Tat ca RL agents	DAT
Epsilon decay	1.0 -> 0.05, decay=0.999993	DAT
Eval mode e=0	set_eval_mode()	DAT
Save/Load	np.savez / np.load	DAT
get_policy()	argmax(Q, axis=1)	DAT

Ket qua evaluation (v2 — 20K episodes)

Agent	Profit (mean +- std)	Stockout Rate	Avg Inventory	Success Rate
Random	232.5 +- 114.3	32.4%	4.35	95.0%
AlwaysOrder2	209.6 +- 57.2	46.7%	2.03	100%
Reorder Threshold	402.1 +- 97.0	16.1%	3.56	100%
Q-Learning	321.1 +- 58.3	38.3%	1.56	100%

Agent	Profit (mean +- std)	Stockout Rate	Avg Inventory	Success Rate
SARSA	339.6 +- 69.9	39.4%	1.37	100%
Double Q-Learning	328.3 +- 73.3	41.3%	1.34	100%

Phan tich

- DAT: RL > Random Agent (339 > 232, +46%)
- DAT: RL > AlwaysOrder2 (339 > 209, +62%)
- DAT: "RL co profit trung binh cao hon always-order baseline" (yeu cau de bai)
- CHUA DAT: RL < Reorder Threshold (339 < 402)

Tai sao RL chua thang Reorder Threshold:

1. **State space lon:** 2,646 states x 6 actions = 15,876 Q-table entries. Voi 20,000 episodes x 30 steps = 600,000 updates, trung binh moi state-action chi duoc cap nhat ~38 lan. Nhieu state it gap (vi du: inventory=20 + regime=high) co the chua duoc kham pha du.
1. **Heuristic don gian nhung hieu qua:** Reorder Threshold (inv<5 -> order 5) truc tiep giai quyét stockout — la van de lon nhat cua bai toan. Day la mot "hard-coded optimal insight" ma RL phai tu hoc.
1. **Demand stochastic:** Regime transitions tao them uncertainty. RL can nhieu data hon de hoc tung regime riêng.
1. **Double Q bi chia doi data:** Moi bang Q chi nhan 50% updates -> can gap doi so episodes de dat cung muc hoc.
1. **Cai thien tu v1:** Profit tang 7-15% so voi 8K episodes, chung to RL dang hoc tot hon khi co nhieu data.

Ghi chu cho bao cao: Thay cho phep giai thich "vi sao chua tot hon" — day la mot phan quan trong cua bao cao phan tich.

iv. Danh gia nhieu seed, baseline ro (20%) -> 18/20 (tang tu 14)

DA DAT

Yeu cau	Trang thai
It nhat 10 seed	10 seeds training + evaluation
Mean +- std	Tat ca metrics deu co
Learning curve	reports/figures/learning_curves.png (20K eps)
So sanh 4+ agents	6 agents
Eval voi e=0	set_eval_mode()
Same training budget	20000 eps x 10 seeds x 3 agents
Weekend surge	Unseen pattern evaluation
Success rate	DA BO SUNG — 95-100%
Avg steps	DA BO SUNG — 30.0 +- 0.0
Stockout days (violations)	DA BO SUNG — 4.8-14.0 ngay/episode

Bang metrics day du (YEU CAU CUA THAY)

Agent	Profit	Stockout Rate	Stockout Days	Avg Steps	Success Rate	Holding Cost	Order Freq
Random	232.5+-114.3	32.4%	9.7+-6.1	30.0	95.0%	65.2	83.1%
AlwaysOrder2	209.6+-57.2	46.7%	14.0+-6.5	30.0	100%	30.4	100%
Reorder	402.1+-97.0	16.1%	4.8+-2.9	30.0	100%	53.3	62.0%
Q-Learning	321.1+-58.3	38.3%	11.5+-4.7	30.0	100%	23.3	81.4%
SARSA	339.6+-69.9	39.4%	11.8+-4.0	30.0	100%	20.6	82.4%
Double Q	328.3+-73.3	41.3%	12.4+-3.7	30.0	100%	20.1	81.0%

CON THIEU NHO

Thieu	Huong khac phuc
Box plot so sanh seeds	Them vao plots.py hoac bao cao

v. Demo chay duoc, truc quan, giai thich duoc (15%) -> 12/15 (tang tu 10)

DA DAT

Yeu cau	Trang thai
Dashboard chon demand regime	DAT — Selectbox Low/Medium/High
Hien thi inventory theo ngay	DAT — Line chart
Hien thi action order	DAT — Bar chart Daily Orders
Bang policy	DAT — 21x6 matrix
Policy heatmap	DAT — Blues colormap, bold text
Duong hoc (learning curves)	DAT — Mean +- std
Nut chuyen giua agents	DAT — Selectbox 6 agents
Agent comparison	DAT — Bar charts 4 metrics
Reward tuc thoi	DA BO SUNG — Daily Reward chart (xanh/do)
Dashboard da verify	DAT — Chay thanh cong
Giao dien chuyen nghiep	DAT — Unified indigo/slate palette

CON THIEU

Thieu	Muc anh huong	Huong khac phuc
Video demo	Cao	Quay 2-3 phut, upload YouTube/Drive

vi. Ma sach, bao cao ro, tai lap duoc (5%) -> 3/5

DA DAT

Yeu cau	Trang thai
Cau truc thu muc chuan	DAT
README.md day du	DAT
requirements.txt	DAT
Docstrings Google style	DAT
Seed reproducibility	DAT
configs.yaml	DAT

CON THIEU

Thieu	Huong khac phuc
reports/final_report.pdf	Viet bao cao 7 phan theo cau truc Thay yeu cau
Link public GitHub	Push + share link

3. Checklist nghiem thu cua Thay

Cau hoi	Tra loi	Danh gia
State co du thong tin?	DAT — inventory + regime + dow + pending	Tot
Invalid action xu ly the nao?	Khong co invalid (0-5 deu hop le). Ngoai range -> assert	Tot
Reward khyen khich dung?	Phan lon dung. Stockout penalty co the tang de RL hoc tot hon	Chap nhan
Terminal xu ly dung trong Q update?	DAT — if done: td_target = reward	Tot

Cau hoi	Tra loi	Danh gia
Agent hoc tot hon random?	DAT — SARSA 339 > Random 232 (+46%)	Tot
Agent tot hon heuristic?	Chua — SARSA 339 < Reorder 402. Da giai thich ly do	Chap nhan
Ket qua on dinh 10 seed?	Std 58-73 cho profit. Chap nhan duoc voi demand stochastic	Chap nhan
Demo the hien policy?	DAT — Heatmap + Policy Table + Daily charts	Tot

4. Cac cai thien da thuc hien (v1 -> v2)

4.1 Dashboard UI (da hoan thanh)

- Bo toan bo emoji lon xon -> giao dien sach
- Thong nhat color palette: Indigo/Slate (chuyen nghiep)
- Chart colors dong bo: xanh la (inventory), xanh duong (orders), cam (demand), do (stockout)
- Sidebar background: Slate-50
- Them Daily Reward chart (instant reward moi ngay, xanh=positive, do=negative)
- Cumulative Reward va Daily Reward hien thi canh nhau (2 columns)
- Policy heatmap chuyen sang Blues colormap, text in dam

4.2 Metrics bo sung (da hoan thanh)

- **Success Rate:** % episodes co profit > 0
- **Avg Steps:** Do dai episode trung binh
- **Stockout Days:** So ngay bi chay hang (constraint violations)
- Hien thi day du trong evaluate.py output

4.3 RL Performance (da cai thien)

- Tang training: 8,000 -> 20,000 episodes
- Epsilon decay cham hon: 0.99998 -> 0.999993
- Eval interval: 400 -> 500
- Ket qua: Profit tang 7-15% cho tat ca RL agents

5. Ke hoach cai thien tiep theo

Buoc	Cong viec	Thoi gian	Diem tang
1	Viet final_report.pdf (7 phan theo yeu cau Thay)	4-6 gio	+2
2	Quay video demo Dashboard	30 phut	+2
3	Push GitHub + them link	15 phut	+1
4	(Tuy chon) Tang STOCKOUT_COST 3->5, re-train	3-5 gio	+2-3

Diem uoc tinh sau cai thien: $87 + 5 = \sim 92/100$

6. Lien ket cheo

- [[Workflow/Training]] — Quy trinh training
- [[Workflow/Evaluation]] — Quy trinh evaluate
- [[Workflow/Dashboard]] — Huong dan chay Dashboard
- [[Error/ERR-001]] — Loi UnicodeEncodeError da fix
- [[Notes/QLearning_Algorithm]] — Ly thuyet Q-Learning
- [[Notes/SARSA_Algorithm]] — Ly thuyet SARSA

- [\[\[Notes/DoubleQLearning_Algorithm\]\]](#) — Ly thuyet Double Q-Learning
- [\[\[Log/2026-05-27\]\]](#) — Log cong viec hom nay